Program #1

Purpose

Demonstrate the ability to create and execute a C++ program using moderately complex control structures and Input/Output routines.

Assignment

Write a program that creates a series of one or more loan repayment schedules. The program should query the user for an input file name and an output file name. The input file will contain the number of schedules that need to be generated as well as the information needed to generate each schedule (initial loan principal, interest percentage rate, and monthly payment). Each generated repayment schedule should be well formatted and should include a breakdown of the principal and interest for each payment, the total number of monthly payments, and the total interest paid for the life of each loan. This information should be printed to the screen as well as sent to the output file.

Each line of the output should show the payment number, the total payment amount, the amount of the total payment that is allocated to interest, the amount of the total payment that is allocated to principal, and the new balance. The sum of the interest amount and principal amount should equal the total payment amount. The last payment may not be exactly the same as the rest. Make sure you check for this.

The program should use a separate function to calculate the repayment table(s). The separate function to calculate the repayment table MUST be in a different input file from the main() routine. See ***Source File Format*** below.

Deliverables

You must submit your homework through ELearning. You must include your .cpp files, your .h file, your input file, and your output file generated from your input file. Code must be well formatted and well commented.

Notes

Remember we are talking about money. All values must represent dollars and cents. You might consider representing all dollar amounts as an integer that represents the number of pennies.

No late homework is accepted.

Input File Format

The input file is a text file that contains integers and doubles. Each number is on its own line. You can create this file using notepad or your favorite text editor. The first entry is an integer that defines the number of schedules that need to be generated using the data in the file. Each set of three lines after the first line will contain the data for a single payment schedule. In order, they will be the principal, annual interest rate, and the monthly payment.

An example input file would look like:

```
3
1000.00
0.035
120.00
5000.00
0.04
165.00
200000.00
0.055
2250.00
```

Example Output Format of a Single Table (screen and file)[1]

```
Starting Balance: $1000.00
Annual Interest Rate: 0.035
Monthly Payment: $120.00

Payment     Amount            Interest    Principal         Balance

1           120.00            2.91        117.09            882.91
2           120.00            2.57        117.43            765.48
3           120.00            2.23        117.77            647.71
4           120.00            1.88        118.12            529.59
5           120.00            1.54        118.46            411.13
6           120.00            1.19        118.81            292.32
7           120.00            0.85        119.15            173.17
8           120.00            0.50        119.50            53.67
9           53.82             0.15        53.67             0.00

Total Payments Made: 9
Total Interest Paid: $13.82
```

[1] Only the first of the three tables is shown here

Source File Format

Your project MUST consist of two independent C++ Source files. The first, at a minimum, contains the main() routine and MUST be named **cs1337F19p1-main.cpp**. The second, at a minimum, contains the routine that calculates and displays the amortization table. It MUST be named **cs1337F19p1-amortize.cpp**. When you have multiple input files, you will also need a header file. This file should be named **cs1337F19p1.h**

Each file must contain a header (the absolute first lines of code) that looks like:

```
//============================================================================
// Name        : <Put File Name Here>
// Author      : <Put your Name here>
// Course      : Put your Course, section, semester, and year like: UTDallas CS 1337.502 F19
// Version     : <Give it a version of your choice>
// Copyright   : <Put the year in here>
//
// Description:
//
//             Write a multiline comment here that describes the contents
//             of this file.
//
//============================================================================
```

Recommended Phases

One way to approach the implementation of this assignment is through incremental development. Coding your project in these incremental "phases" may help with the understanding and organization of your code. You are not required to utilize these phases.

Phase 1

Create a working "Hello World" Project in Eclipse

Phase 2

Prompt the user for a file name and then print the file name to the screen, Exit the program.

Phase 3

Using the provided file names, open the input and output files, close them, and then exit.

Phase 4

Using the provided file names, open the input and output files. Read the first integer in the file and print to screen and to the output file. Exit. Verify screen and file output match contents of the input file's first integer.

Phase 5

Open the file and read the first integer. Using a loop, read the rest of the values from the input file and print to screen and to the output file. Exit. Verify screen and file output match contents of the input file.

Phase 6

From within the loop, once you have read a full set of input values for a single loan, call a function and pass these values as parameters to the function. FROM WITHIN the function, print the parameter values to the screen and output file. Exit. Verify screen and file output match what you expect.

Phase 7

Flesh out the function to print the amortization table for the loan values passed in. This is where you will have to UNDERSTAND how to create the table in your head before you start coding C++. If you do not know how to generate a table given the input values, you will not be able to generate code to print the table.